

Report Period	Sep 15, 2026 10:43 - Sep 15, 2026 10:48
Generated At	2026-09-25 14:18:29
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		13906	
Total Threats Mitigated	13	Total Packets Analyzed		3991576877	
High Priority	9	Malicious Packets		3983726076	
Low Priority	4	True Negatives Passed		10151	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	0	UDP Flood		10	
FP Rate	0.00%	Uncertain		3	
IF Precision	100.00%				
IF Recall	98.70%	Action		Count	
IF F1-Score	99.35%	Quarantined		3	
IF Accuracy	99.65%	Time Ban		3	
		Blackhole		0	
		Released		1	

Traffic Summary	Count
Total Flows Observed	13906
Total Packets Analyzed	3991576877
Malicious Packets	3983726076
True Negatives Passed	10151
Attack Vector	Count
ICMP Flood	0
SYN Flood	0
UDP Flood	10
Uncertain	3
Action	Count
Quarantined	3
Time Ban	3
Blackhole	0
Released	1

Metric	Value
Total Threats Mitigated	13
High Priority	9
Low Priority	4
Manual Releases	0
Manual Blocks	0
ML Metrics	Value
False Positives	0
FP Rate	0.00%
IF Precision	100.00%
IF Recall	98.70%
IF F1-Score	99.35%
IF Accuracy	99.65%

Attack Vector	Count
ICMP Flood	0
SYN Flood	0
UDP Flood	10
Uncertain	3
Action	Count
Quarantined	3
Time Ban	3
Blackhole	0
Released	1

ML Metrics	Value
False Positives	0
FP Rate	0.00%
IF Precision	100.00%
IF Recall	98.70%
IF F1-Score	99.35%
IF Accuracy	99.65%

Action	Count
Quarantined	3
Time Ban	3
Blackhole	0
Released	1

Released	1
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2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.70%	Share of actual attacks successfully flagged
F1-Score	99.35%	Balanced measure combining Precision and Recall
Accuracy	99.65%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.30%	Actual attacks that went undetected
True Positive Rate (TPR)	98.70%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3727	FN 49
Actual: Normal	FP 0	TN 10102

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	3727

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.70%	Share of actual attacks successfully flagged
F1-Score	99.35%	Balanced measure combining Precision and Recall
Accuracy	99.65%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.30%	Actual attacks that went undetected
True Positive Rate (TPR)	98.70%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3727	FN 49
Actual: Normal	FP 0	TN 10102

2d. Response Latency

Metric	Value	Description
Detection Time	284.7 ms	Average time taken to detect an attack
Mitigation Response Time	8.7 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.73%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	1.62%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.90%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.22	3	3	2026-09-15 10:45:57	2026-09-15 10:47:19
10.0.0.20	2	2	2026-09-15 10:45:56	2026-09-15 10:47:07
10.0.0.21	2	2	2026-09-15 10:45:56	2026-09-15 10:47:07

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 10:45:56	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:45:56	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:45:57	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:46:07	10.0.0.21	DDoS	UDP Flood	75.9%	High	Time Ban (1m)
2026-09-15 10:46:07	10.0.0.20	DDoS	UDP Flood	84.5%	High	Time Ban (1m)
2026-09-15 10:46:07	10.0.0.21	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-15 10:46:07	10.0.0.20	DDoS	UDP Flood	84.5%	High	Time Ban
2026-09-15 10:46:08	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-15 10:46:08	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-15 10:47:07	10.0.0.21	DDoS	UDP Flood	79.0%	High	Time Ban (2m)
2026-09-15 10:47:07	10.0.0.20	DDoS	UDP Flood	84.5%	High	Time Ban (2m)
2026-09-15 10:47:08	10.0.0.22	DDoS	UDP Flood	85.9%	High	Released
2026-09-15 10:47:19	10.0.0.22	DDoS	UDP Flood	85.9%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6402	85.9%	Critical	Phase 2	3	11s	2026-09-15 10:47:19	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6132	85.9%	High	Phase 2	1	1m 11s	2026-09-15 10:47:08	Ban Expired
10.0.0.20	UDP Flood	0.6167	84.5%	High	Phase 2	2	1m 11s	2026-09-15 10:47:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6399	79.0%	High	Phase 2	2	1m 11s	2026-09-15 10:47:07	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	77.7%	High	Phase 2	1	11s	2026-09-15 10:46:08	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	84.5%	High	Phase 2	1	11s	2026-09-15 10:46:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	75.9%	High	Phase 2	1	11s	2026-09-15 10:46:07	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE